Final exam · Unit 4 · Nonlinear systems and stability · 2.5 points · 50 min · Standard exam level

P17 — Predators and prey with logistic prey growth: equilibria, Jacobians, the focus-node threshold of the coexistence point and the long-time outcome

After scaling, the densities $x(t) \geq 0$ of a prey and $y(t) \geq 0$ of its predator obey

$$\dot{x} = x(1-x) - xy, \qquad \dot{y} = y\,(x-a),$$

where the parameter a, with $0 < a < 1$, is the death rate of the predator measured against its rate of conversion of prey. Without predators the prey grows logistically with carrying capacity 1.

(a) Find all the equilibria in the closed first quadrant and the Jacobian matrix of the system. Classify the two equilibria that lie on the axes and give their eigen-directions.

(b) Show that the coexistence equilibrium is asymptotically stable for every $a \in (0,1)$, and find the value a^* that separates a stable focus from a stable node. Describe the phase portrait near this point for $a = \frac{1}{2}$ (eigenvalues, sense of rotation) and for $a = \frac{9}{11}$ (eigenvalues, eigenvectors, the change of basis P with $P^{-1}JP$, and the slow direction).

(c) Describe the global phase portrait in the open quadrant (nullclines and the direction of the flow) and state the ecological long-time outcome. Hint for the global part: study $V(x,y) = x - a - a\ln\dfrac{x}{a} + y - b - b\ln\dfrac{y}{b}$ with $b = 1-a$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns $x(t), y(t)$, independent variable $t \geq 0$; state space the closed quadrant $x \geq 0, y \geq 0$ (densities).
- Autonomous, nonlinear (quadratic) planar system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ with $\mathbf{f}$ polynomial, hence $\mathbf{f}$ and $D\mathbf{f}$ are continuous: the existence and uniqueness theorem for systems (Robinson Thm 25.2) gives one solution through each point, and orbits cannot cross.
- The axes are invariant: if $x = 0$ then $\dot{x} = 0$; if $y = 0$ then $\dot{y} = 0$. Hence the open quadrant is invariant and densities never become negative.

Symbol	Meaning	Range
x	prey density (scaled by the carrying capacity)	$x \geq 0$
y	predator density (scaled)	$y \geq 0$
a	predator death rate / conversion rate	$0 < a < 1$

Symbol	Meaning	Range
$b = 1 - a$	predator density at coexistence	$0 < b < 1$

Phase 2. Canonical formulation

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \qquad \mathbf{f}(x, y) = \begin{pmatrix} x - x^2 - xy \\ xy - ay \end{pmatrix}.$$

Near an equilibrium $\mathbf{x}^*$, with $\boldsymbol{\xi} = \mathbf{x} - \mathbf{x}^*$, the linearised system is $\dot{\boldsymbol{\xi}} = J(\mathbf{x}^*)\,\boldsymbol{\xi}$ (Robinson Sec. 32.4.2). When no eigenvalue of $J(\mathbf{x}^*)$ has zero real part, the Hartman-Grobman theorem (Robinson Sec. 32.4.3) says the nonlinear portrait near $\mathbf{x}^*$ is the linear one, slightly deformed.

Part (a) · 0.8 points

Equilibria, Jacobian and the boundary equilibria

Step 1. Equilibria

$\dot{x} = 0 \iff x = 0$ or $y = 1 - x$; $\dot{y} = 0 \iff y = 0$ or $x = a$. Combine one condition from each equation:

- $x = 0$ and $y = 0$: $O = (0, 0)$.
- $x = 0$ and $x = a$: impossible, since $a > 0$.
- $y = 1 - x$ and $y = 0$: $x = 1$, so $E_1 = (1, 0)$.
- $y = 1 - x$ and $x = a$: $E^* = (a, 1 - a)$, inside the open quadrant because $0 < a < 1$.

Step 2. Jacobian

Differentiate each component with respect to each variable:

$$\frac{\partial}{\partial x}(x - x^2 - xy) = 1 - 2x - y, \quad \frac{\partial}{\partial y}(x - x^2 - xy) = -x, \quad \frac{\partial}{\partial x}(xy - ay) = y, \quad \frac{\partial}{\partial y}(xy - ay) = x - a,$$

$$J(x, y) = \begin{pmatrix} 1 - 2x - y & -x \\ y & x - a \end{pmatrix}.$$

Step 3. The origin

$$J(0,0) = \begin{pmatrix} 1 & 0 \\ 0 & -a \end{pmatrix}.$$

The matrix is diagonal, so the eigenvalues are $\lambda_1 = 1$ with $\mathbf{v}_1 = (1,0)$ and $\lambda_2 = -a < 0$ with $\mathbf{v}_2 = (0,1)$ (Robinson Example 27.2). Opposite signs: a **saddle** (Robinson Sec. 28.3), hyperbolic, so the nonlinear point is a saddle too. The x-axis is its unstable direction (prey alone grow), the y-axis its stable direction (predators alone die out as $y = y_0 e^{-at}$, which is exact because the y-axis is invariant).

Step 4. The prey-only equilibrium

$$J(1,0) = \begin{pmatrix} 1-2-0 & -1 \\ 0 & 1-a \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 0 & 1-a \end{pmatrix}.$$

It is upper triangular, so its eigenvalues are the diagonal entries (Robinson Example 27.3): $\lambda_1 = -1$ and $\lambda_2 = 1 - a > 0$. Eigenvectors from $(J - \lambda I)\mathbf{v} = \mathbf{0}$:

- $\lambda_1 = -1$: $\begin{pmatrix} 0 & -1 \\ 0 & 2-a \end{pmatrix}\mathbf{v} = \mathbf{0}$. Row 1 gives $-v_2 = 0$, so $\mathbf{v}_1 = (1,0)$.
- $\lambda_2 = 1 - a$: $\begin{pmatrix} -(2-a) & -1 \\ 0 & 0 \end{pmatrix}\mathbf{v} = \mathbf{0}$. Row 1 gives $-(2-a)v_1 - v_2 = 0$, so $v_2 = (a-2)v_1$ and $\mathbf{v}_2 = (1, a-2)$.

Opposite signs: E_1 is a **saddle**. Its unstable direction $\pm(1, a-2)$ enters the open quadrant along $(-1,\ 2-a)$, up and to the left: a small number of predators introduced into a prey population at capacity grows.

KEY RESULTS

- Equilibria: $O = (0,0)$, $E_1 = (1,0)$, $E^* = (a,\ 1-a)$.
- $J(x,y) = \begin{pmatrix} 1-2x-y & -x \\ y & x-a \end{pmatrix}$.
- O: eigenvalues 1 (along $(1,0)$) and $-a$ (along $(0,1)$), saddle.
- E_1: eigenvalues -1 (along $(1,0)$) and $1-a>0$ (along $(1, a-2)$), saddle.

Part (b) · 1 point

Stability of the coexistence point and the focus-node threshold

Step 1. Jacobian at the coexistence point

Substitute $x = a, y = 1 - a$ in J: $1 - 2a - (1 - a) = -a$, $-x = -a$, $y = 1 - a$, $x - a = 0$:

$$J^* = J(a, 1 - a) = \begin{pmatrix} -a & -a \\ 1 - a & 0 \end{pmatrix}.$$

Step 2. Characteristic polynomial

$$\det(J^* - \lambda I) = \begin{vmatrix} -a - \lambda & -a \\ 1 - a & -\lambda \end{vmatrix} = (-a - \lambda)(-\lambda) - (-a)(1 - a) = \lambda^2 + a\lambda + a(1 - a).$$

So $\operatorname{tr} J^* = -a$, $\det J^* = a(1 - a)$ and

$$\lambda = \frac{-a \pm \sqrt{a^2 - 4a(1 - a)}}{2} = \frac{-a \pm \sqrt{a(5a - 4)}}{2}.$$

Step 3. Stability for every $a \in (0, 1)$

$\det J^* = a(1 - a) > 0$ excludes a saddle, and $\operatorname{tr} J^* = -a < 0$ gives both eigenvalues negative real part: if they are real, their product is positive and their sum negative, so both are negative; if they are complex, the real part is $-a/2 < 0$. Every eigenvalue has nonzero real part, so the point is hyperbolic and by Hartman-Grobman (Robinson Sec. 32.4.3) E^* is **asymptotically stable** in the nonlinear system.

Step 4. The threshold

The type depends on the sign of $\Delta(a) = a(5a - 4)$ (Robinson Ch. 31, summary of phase portraits):

- $0 < a < \frac{4}{5}$: $\Delta < 0$, complex pair with real part $-a/2$: **stable focus** (spiral).
- $a = \frac{4}{5}$: $\Delta = 0$, double eigenvalue $\lambda = -\frac{2}{5}$; $J^* - \lambda I = \begin{pmatrix} -\frac{2}{5} & -\frac{4}{5} \\ \frac{1}{5} & \frac{2}{5} \end{pmatrix} \neq 0$, so there is a single eigen-direction, $\mathbf{v} = (2, -1)$ from row 1: **stable degenerate (improper) node** (Robinson Ch. 30). The eigenvalue is still negative, so the point stays hyperbolic and asymptotically stable; only the fine shape (node or slow spiral) may be changed by the nonlinear terms in this borderline case.
- $\frac{4}{5} < a < 1$: $\Delta > 0$, two distinct negative eigenvalues: **stable node**.

So $a^* = \frac{4}{5}$: predators that die fast (large a) let the system settle without oscillating.

Step 5. Focus case $a = \frac{1}{2}$

$\lambda^2 + \frac{1}{2}\lambda + \frac{1}{4} = 0$, so $\lambda = \dfrac{-\frac{1}{2} \pm \sqrt{\frac{1}{4} - 1}}{2} = -\dfrac{1}{4} \pm \dfrac{\sqrt{3}}{4}i$. Real part $-\frac{1}{4}$ (decay rate), imaginary part $\frac{\sqrt{3}}{4}$ (angular frequency; period of the linear oscillation $2\pi/(\sqrt{3}/4) = 8\pi/\sqrt{3}$).

Sense of rotation from a test vector (Robinson Ch. 29): at $\boldsymbol{\xi} = (1, 0)$ (more prey than at equilibrium), $\dot{\boldsymbol{\xi}} = J^*\boldsymbol{\xi} = (-\frac{1}{2},\ \frac{1}{2})$. The second component is positive: the orbit turns from the positive ξ_1 axis towards the positive ξ_2 axis, i.e. **counter-clockwise** in the (x, y) plane. Ecologically: an excess of prey makes the predators grow, the predators then eat the prey down, and so on, with oscillations of decreasing amplitude.

Step 6. Node case $a = \frac{9}{11}$

$$J^* = \begin{pmatrix} -\frac{9}{11} & -\frac{9}{11} \\ \frac{2}{11} & 0 \end{pmatrix}, \qquad \lambda^2 + \frac{9}{11}\lambda + \frac{9}{11}\cdot\frac{2}{11} = 0, \qquad \lambda = \frac{-\frac{9}{11} \pm \sqrt{\frac{81}{121} - \frac{72}{121}}}{2} = \frac{-\frac{9}{11} \pm \frac{3}{11}}{2}.$$

So $\lambda_1 = -\frac{3}{11}$ and $\lambda_2 = -\frac{6}{11}$ (check: sum $-\frac{9}{11} = \mathrm{tr}$, product $\frac{18}{121} = \det$). Eigenvectors from the first row of $(J^* - \lambda I)\mathbf{v} = \mathbf{0}$ (the second row is proportional because $\det(J^* - \lambda I) = 0$):

- $\lambda_1 = -\frac{3}{11}$: $\left(-\frac{9}{11} + \frac{3}{11}\right) v_1 - \frac{9}{11}v_2 = 0 \Rightarrow -6v_1 - 9v_2 = 0 \Rightarrow \mathbf{v}_1 = (3, -2)$. Second row check: $\frac{2}{11}\cdot 3 + \frac{3}{11}\cdot(-2) = 0$.
- $\lambda_2 = -\frac{6}{11}$: $\left(-\frac{9}{11} + \frac{6}{11}\right) v_1 - \frac{9}{11}v_2 = 0 \Rightarrow -3v_1 - 9v_2 = 0 \Rightarrow \mathbf{v}_2 = (3, -1)$. Second row check: $\frac{2}{11}\cdot 3 + \frac{6}{11}\cdot(-1) = 0$.

Change of basis (Robinson Sec. 28.2): $P = [\mathbf{v}_1 \mid \mathbf{v}_2] = \begin{pmatrix} 3 & 3 \\ -2 & -1 \end{pmatrix}$, $\det P = 3(-1) - 3(-2) = 3$, and

$$P^{-1} = \frac{1}{3}\begin{pmatrix} -1 & -3 \\ 2 & 3 \end{pmatrix}.$$

$J^*P = [\lambda_1\mathbf{v}_1 \mid \lambda_2\mathbf{v}_2] = \begin{pmatrix} -\frac{9}{11} & -\frac{18}{11} \\ \frac{6}{11} & \frac{6}{11} \end{pmatrix}$, and entry by entry

$$\begin{aligned}
(P^{-1}J^*P)_{11} &= \tfrac{1}{3}\left[(-1)(-\tfrac{9}{11}) + (-3)\tfrac{6}{11}\right] = \tfrac{1}{3}\cdot\left(-\tfrac{9}{11}\right) = -\tfrac{3}{11}, \\
(P^{-1}J^*P)_{12} &= \tfrac{1}{3}\left[(-1)(-\tfrac{18}{11}) + (-3)\tfrac{6}{11}\right] = 0, \\
(P^{-1}J^*P)_{21} &= \tfrac{1}{3}\left[2(-\tfrac{9}{11}) + 3\cdot\tfrac{6}{11}\right] = 0, \\
(P^{-1}J^*P)_{22} &= \tfrac{1}{3}\left[2(-\tfrac{18}{11}) + 3\cdot\tfrac{6}{11}\right] = -\tfrac{6}{11}.
\end{aligned}$$

So $P^{-1}J^*P = \mathrm{diag}(-\frac{3}{11}, -\frac{6}{11})$. In $\mathbf{z} = P^{-1}\boldsymbol{\xi}$ coordinates $z_1 = c_1 e^{-3t/11}$, $z_2 = c_2 e^{-6t/11}$, and eliminating t, $z_2 = K z_1^2$: parabola-like orbits. All orbits enter E^* tangent to the **slow** eigenvector $\mathbf{v}_1 = (3, -2)$ (smaller $|\lambda|$), and far from E^* they run parallel to the fast one $\mathbf{v}_2 = (3, -1)$ (Robinson Sec. 28.3-28.4).

KEY RESULTS

- $J(E^*) = \begin{pmatrix} -a & -a \\ 1-a & 0 \end{pmatrix}$, $\mathrm{tr} = -a < 0$, $\det = a(1-a) > 0$: asymptotically stable for all $a \in (0, 1)$.
- Discriminant $a(5a-4)$: stable focus for $0 < a < \frac{4}{5}$, stable degenerate node at $a^* = \frac{4}{5}$, stable node for $\frac{4}{5} < a < 1$.
- $a = \frac{1}{2}$: $\lambda = -\frac{1}{4} \pm \frac{\sqrt{3}}{4} i$, counter-clockwise spiral.
- $a = \frac{9}{11}$: $\lambda_1 = -\frac{3}{11}$, $\mathbf{v}_1 = (3, -2)$ (slow), $\lambda_2 = -\frac{6}{11}$, $\mathbf{v}_2 = (3, -1)$ (fast); $P^{-1}JP = \mathrm{diag}(-\frac{3}{11}, -\frac{6}{11})$.

Part (c) · 0.7 points

Global portrait and long-time outcome

Step 1. Nullclines and direction field

In the open quadrant, $\dot{x} > 0$ below the line $x + y = 1$ and $\dot{x} < 0$ above it; $\dot{y} > 0$ to the right of $x = a$ and $\dot{y} < 0$ to the left. The two lines cross at E^* and cut the quadrant into four regions; the flow goes right-and-up (below the line, right of $x = a$), left-and-up (above, right), left-and-down (above, left) and right-and-down (below, left). Going round these four regions in that order is a counter-clockwise turn about E^*, consistent with (b). The axes are invariant: on the x axis orbits run to E_1, on the y axis to O.

Step 2. The function V (extension)

A **Lyapunov function** (extension: not in the course; a C^1 function with a strict minimum at the equilibrium that does not increase along orbits) proves global statements that linearisation cannot. Here V is defined in the open quadrant. Its derivatives are

$$V_x = 1 - \frac{a}{x} = \frac{x-a}{x}, \qquad V_y = 1 - \frac{b}{y} = \frac{y-b}{y}.$$

By the chain rule, $\frac{d}{dt} V(x(t), y(t)) = V_x \dot{x} + V_y \dot{y}$:

$$\dot{V} = \frac{x-a}{x}\, x(1-x-y) + \frac{y-b}{y}\, y(x-a) = (x-a)\big[(1-x-y) + (y-b)\big] = (x-a)(1-b-x).$$

With $1 - b = a$: $\dot{V} = (x-a)(a-x) = -(x-a)^2 \le 0$.

Also $V \geq 0$ with equality only at E^*: the function $g(s) = s - c - c\ln(s/c)$ has $g'(s) = 1 - c/s$, negative for $s < c$ and positive for $s > c$, so its minimum is $g(c) = 0$; V is the sum of $g(x)$ with $c = a$ and $g(y)$ with $c = b$. $V \to \infty$ at the axes and at infinity, so every level set $V = \text{const}$ is a closed curve around E^* and orbits stay bounded.

Step 3. Long-time outcome

V decreases strictly except where $x = a$. On the line $x = a$ away from E^*, $\dot{x} = a(1 - a - y) \neq 0$, so the orbit leaves the line at once: no orbit other than E^* stays where $\dot{V} = 0$. Hence (LaSalle's invariance principle, extension) $V(x(t), y(t)) \to 0$ and every orbit starting in the open quadrant tends to E^*. The linearisation of (b) gives the final approach: spirals for $a < \frac{4}{5}$, tangent to the slow eigenvector for $a > \frac{4}{5}$.

Ecological outcome: neither species dies out; prey and predator settle to $x = a$, $y = 1 - a$. The prey density at coexistence equals the predator's death parameter; a less efficient predator (larger a) leaves more prey and fewer predators, and for $a \geq 1$ (outside the data) E^* leaves the quadrant and the predator goes extinct.

KEY RESULTS

- Nullclines: $\dot{x} = 0$ on $x = 0$ and on $x + y = 1$; $\dot{y} = 0$ on $y = 0$ and on $x = a$.
- $\dot{V} = -(x - a)^2 \leq 0$ (extension: Lyapunov function).
- Every orbit with $x(0) > 0, y(0) > 0$ tends to $E^* = (a, 1 - a)$: prey and predators coexist; with $a = \frac{1}{2}$ through damped oscillations, with $a = \frac{9}{11}$ without them.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $a \to 0^+$: $E^* \to (0, 1)$, trace $\to 0$, weakly damped oscillations (close to a centre). $a \to 1^-$: $E^* \to E_1 = (1, 0)$ and the eigenvalue $1 - a$ of E_1 tends to 0: the two equilibria merge (transcritical bifurcation, extension). For all $a \in (0, 1)$, $t \to \infty$ gives $(x, y) \to (a, 1 - a)$.
- **Checks:** $\mathbf{f}(a, 1 - a) = \big(a(1 - a - (1 - a)),\ (1 - a)(a - a)\big) = \mathbf{0}$; tr and det match sums and products of the eigenvalues in both cases; $P^{-1}J^*P$ is diagonal; numerical integration from $(0.2, 0.2)$ ends at E^* for $a = \frac{1}{2}$ and $a = \frac{9}{11}$ (script). The model is dimensionless, so no units appear.
- **Common mistakes in the exam:** (1) writing the classical Lotka-Volterra centre: the logistic term $-x^2$ adds $-a$ to the trace and makes E^* a sink; (2) concluding "stable" from $\det > 0$ alone, without the sign of the trace; (3) taking the fast eigenvector as the direction of approach of a node: orbits arrive tangent to the slow one, $(3, -2)$; (4) using the linearisation to claim global convergence: Hartman-Grobman is only local, the global statement needs V.